

GLOBAL SALES & OPERATIONS VISUAL ANALYTICS

Technical Dashboard Report

Dataset: Northwind Traders

Tool Used: Tableau

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This Report Covers Across the Following:

1. Introduction and Objective
2. Dataset Overview
3. Dashboard Structure
4. Page 1: Sales & Revenue Overview
5. Page 2: Product & Category
6. Page 3: Operational & People Performance
7. Page 4: Customer Experience
8. Key Business Findings
9. Limitations and Recommendations

INTRODUCTION AND OBJECTIVE

This report documents the development of an interactive business intelligence dashboard using the Northwind Traders dataset. The objective of this dashboard is to visually explain global sales performance, product contribution, regional distribution, operational efficiency, and salesforce productivity.

The dashboard combines information from customers, orders, order details, products, employees, and shippers into one view. The solution is divided into four pages, each addressing a specific business focus area.

The dashboard supports interactive filtering and allows evaluation across time, geography, product categories, and operational dimensions.

DATASET OVERVIEW

The Northwind dataset contains transactional and master data across seven business aspects.

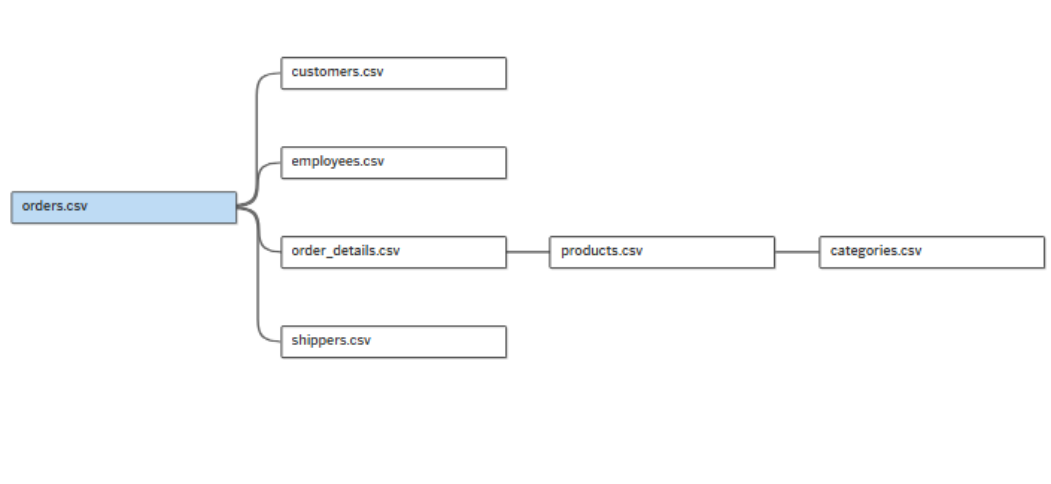
Tables Used

- Customers
- Orders
- Order Details
- Products
- Categories
- Employees
- Shippers

Key Metrics Created

- Actual Total Price ($[Unit\ Price] * [Quantity] * (1 - [Discount])$)
- Total Discount ($[Total\ Price] * [Discount]$)
- Shipping Days ($ROUND(DATEDIFF('day', [Order\ Date], [Shipped\ Date]), 0)$)
- Rank of Products ($RANK(SUM([Total\ Price]))$)

This is what the data looks like in one picture:



3. DASHBOARD STRUCTURE

The dashboard is divided into four pages:

Page 1 — Sales & Revenue Overview

Page 2 — Product & Category Intelligence

Page 3 — Operational & People Performance

Page 4 — Customer Experience

Interactive dashboard actions enable cross-filtering; selecting any data point dynamically updates all related visualizations for granular analysis.

PAGE 1 — SALES & REVENUE OVERVIEW

Objective

To analyze revenue trends, order volume, and seasonal patterns.

Visual Components

- Revenue growth over time
- Order volume trend
- Revenue by year
- Revenue by location

Analysis

Revenue and order volume were plotted over time using order date. Both measures show variation across months, indicating demand fluctuation. Periods with higher order volume relate to higher revenue totals.

Seasonal behavior is visible in repeated increases across similar months. This indicates recurring purchasing cycles.

The map clearly shows that most of the revenue comes from USA followed closely by Brazil and Canada, generating up to \$245,584, \$106,925, \$50,196 respectively.

PAGE 2 — PRODUCT & CATEGORY INTELLIGENCE

Objective

To identify top performing product categories and evaluate discontinued products.

Visual Components

- Top products / category by revenue
- Discontinued vs active product comparison
- Discount vs Total Revenue

Analysis

- Category level aggregation shows a hierarchy of the categories by how much income they generate.
- Product-level ranking identifies individual products contributing the largest share of revenue.
- Some discontinued products like meat and Poultry contribute greatly to total income generated across the category. However, no future growth can occur from discontinued items. This indicates potential revenue loss if discontinued products previously generated consistent demand.
- The comparison between discounts and total revenue shows similar pattern, suggesting that more discounts bring more revenue.

PAGE 3 — OPERATIONAL & PEOPLE PERFORMANCE

Objective

To evaluate shipping cost distribution and employee sales contribution.

Visual Components

- Freight cost by country
- Revenue by employee
- Shipper cost and speed comparison

Analysis

- Freight cost comparison shows variation across countries. On the very top are countries that generated the most revenue.
- Employee performance comparison identifies sales contribution per employee across the years.
- Shipper comparison shows cost variation between shipping providers and the average time they take to deliver. The variations show that less money is spent on the fastest deliveries even though it could be due to distance, shorter distances would cost less and get shipped quicker.

PAGE 4 — Customer Experience

Objective

To explain the impact of operations on customers.

Visual Components

- Average shipping time (days)
- Effect of shipping speeds on customers

Analysis

- Discount levels vary across product categories. Higher revenue generating customers get more discounts.
- Revenue increases with moderate discounting. However, high discount levels can affect profits.
- The comparison between revenue and discount shows that not all discounting leads to higher total revenue.

This indicates that discount strategy should be monitored at category level.

KEY BUSINESS FINDINGS

Revenue is concentrated in selected product categories.

Moderate discounting drives revenue growth.

A subset of countries generates the majority of sales.

Freight costs vary across regions and shippers.

Employee sales contribution is uneven.

Discontinued products previously contributed to revenue.

LIMITATIONS AND RECOMMENDATIONS

Limitations

The data tells where the deliveries go but it does not include where they are from limiting measuring the distance covered, this limitation makes it impossible to properly identify why some shipments take long.

Recommendations

Data about warehouse locations should be included to track the distance that orders cover.

Include delivery duration for shipping efficiency analysis.

Track inventory levels for product availability analysis.

Add sales targets for employee performance evaluation.

Review discount strategy at category level.

[Dashboard Link](#)